



Plasma MicroRNA and Protein Signature Associated with Obesity in the Chinese Han Population Using MicroRNA Omics and Proteomics Analysis of 1026 Subjects

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Received: 7 November 2023 / Revised: 31 March 2025 / Accepted: 7 May 2025
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Abstract

Overweight and obesity have become major health risks around the world. The risk factors and mechanisms for overweight and obesity are not fully understood. It is critical to establish reliable factors associated with overweight and obesity. For future predictions, we examined the microRNA (miRNA) and protein expression levels of the Chinese population. Between 2017 and 2021, we conducted a community-based prospective cohort study of 1026 Chinese Han healthy adults in Zhengzhou, China. In this study, we analyzed the differentially expressed miRNA and protein profiles generated from 1026 subjects using RNA sequencing and proteomic analysis to investigate the potential association of levels of differential miRNAs and proteins with overweight and/or obesity in the Chinese Han adults. In our study, we identified 21 differential expression levels as a miRNA signature (10 up-regulated: hsa-miR-199b-5p, hsa-miR-548av-5p, hsa-miR-548 k, hsa-miR-362-5p, hsa-miR-181c-3p, hsa-miR-664b-5p, hsa-miR-148a-5p, hsa-miR-885-5p, hsa-miR-18a-3p, hsa-miR-590-3p and hsa-miR-450b-5p; 11 down-regulated: hsa-miR-26b-3p, hsa-miR-143-3p, hsa-miR-145-5p, hsa-miR-760, hsa-miR-205-5p, hsa-miR-654-5p, hsa-miR-1306-3p, hsa-miR-5010-5p, hsa-miR-655-3p, hsa-miR-196b-5p and hsa-miR-141-3p) associated with overweight or obese subjects. We then identified these potential miRNA direct target proteins and the associated miRNA-protein regulatory network with the Chinese obese phenotype. Furthermore, we found that these miRNA signatures and protein panels are associated with the onset and progression of obesity-related illnesses. These miRNA signatures and target protein panels may be used to predict obesity, identify potential new pathways, and serve as new biomarkers to predict overweight- and obesity-related diseases in the future.

Keywords MicroRNA omics · Proteomics · Overweight · Obesity · Chinese population

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Abbreviations

BMI	Body mass index
GSEA	Gene set enrichment analysis
Log ₂ FC	Log ₂ FoldChange
MS	Mass spectrometry
PTMs	Post-translational modifications
iBAQ	Intensity based absolute quantification
BH	Benjamini & Hochberg
ROC	Receiver operating characteristic

Introduction

In recent years, the rise in overweight and obesity has become a major health concern around the world (Roberto et al. 2015). Obesity has been considered a global epidemic in developed countries by World Health Organization (WHO) since 1998, and several related chronic nutrition disease and obesity have become a public health priority to be solved. As a result, overweight and obesity, as well as chronic non-transmissible diseases, have been included in WHO's plan to reduce prevalence or the global disease burden worldwide in 2025 (Shamah Levy 2016). Overweight and obesity are major issues for society today, with overweight people outnumbering underweight persons in the majority of countries throughout the world (Gomez Puente and Martinez-Marcos 2018). Overweight and obesity are major risk factors for multiple diseases, including cardiovascular disease (CVD), Type 2 Diabetes, musculoskeletal disorders, stroke and development of various cancers, which account for more than 70% of all deaths in persons aged 30–70 (Williams et al. 2015). Obesity tends to cluster in specific families because they share linked genes and comparable eating and exercise behaviors, such as unhealthy diet, liquid calories, and inactivity (Tchang et al. 2021). In addition, energy imbalances between calories consumed and calories expended are a major cause of overweight and obesity globally, which includes an increased intake of energy-dense foods that are high in fat and sugars, and a decrease of physical activity as a result of the increasingly sedentary nature of many forms of work, changing modes of transportation, and increasing urbanization (Conway and Rene 2004). The primary causes of these changes in dietary and physical activity patterns are environmental and societal changes associated with development, as well as lack of supportive policies in sectors such as health, agriculture, transportation, urban planning, environment, food processing, distribution, marketing, and education (Mozaffarian 2016). To further understand the coordinated molecular network regulation system in overweight and obesity, we conducted a combined microRNA (miRNA)-omics and proteomics analysis on peripheral blood samples from a cohort of 1026 Chinese individuals.

Body mass index (BMI) is a weight-for-height indicator used to classify persons as overweight or obese. BMI is calculated by dividing body weight in kilograms by body height in meters squared (kg/m^2). In China, obesity is defined as a BMI of 28 or higher, whereas overweight is a BMI of 24 or higher. BMI value is the most commonly used metric to measure the prevalence of overweight and obesity among persons of all ages (Weir and Jan 2021). Overweight and obesity are both terms used to identify persons who are at risk of developing health problems as a result of excess fat. If overweight is not addressed in a timely manner, it can quickly progress to obesity (Table 1).

MiRNAs are a class of short non-coding RNAs that regulate the expression of target genes at the post-transcriptional level (Bartel 2009). In addition, miRNAs have been used as prognostic, diagnostic, and therapeutic biomarkers in a variety of diseases (Shah and Calin 2014). MiRNAs are frequently clustered in the human genome at diverse positions inside gene sequences or in intronic regions, and they may influence around 30% of gene expression across the whole mammalian genome (Filipowicz et al. 2008). Recently, miRNAs have been shown to influence insulin resistance, adipogenesis, cardiovascular disease, and endothelial dysfunction in overweight and obese individuals (Cirillo et al. 2018; Santulli 2016; Zaiou et al. 2018). The chromosomal regions associated with obesity contain around 79 differentially expressed miRNAs (Shi et al. 2016a, 2016b). MiRNAomics was used to investigate a complex signal network regulated by nitric oxide-mediated miRNA in response to low temperatures and nitric oxide (Xu 2021). Mass spectrometry (MS)-based proteomics is a widely used method for identifying and measuring proteins in samples. Proteomics has been used to investigate whole proteomes, drug screenings, biomarker discovery, target validation, and post-translational modifications (PTMs) (Zhang et al. 2014). The correlations of various microRNAs and their protein targets with obesity and obesity-related disorders in the Chinese population are unknown at this time, and the molecular processes behind these associations have yet to be fully understood. Multi-omics analyses of obesity in the Chinese population have not yet been fully understood. In this study, we utilized miRNA sequencing and proteomics to analyze the peripheral blood samples from 1026 Chinese Han adults to address the following questions: (1) Analyze the obesity status and investigate molecular risk factors of obesity in the Chinese population; (2) Identify specific miRNA signatures that are associated with obesity in the Chinese population; (3) Identify miRNA targets and proteins that are associated with obesity in the Chinese population; (4) Analyze miRNA and/or protein signatures to predict the risk of overweight and obesity in the Chinese population. Our findings provide potential novel mechanisms and biomarkers for predicting and preventing obesity-related disorders in the future.

Table 1 The analysis of factors related to BMI

Characteristic	Number of samples	%	Body Mass Index Classification				χ^2	P value
			Underweight t	Normal	Overweight	Obese		
All sample	1026	100	28	489	414	95		
Sex							28.17	3.35E-06
Male (1)	395	38.5	10	150	184	51		
Female (2)	631	61.5	18	339	230	44		
Age							46.228	5.07E-10
<50	628	61.2	342	50	211	25		
≥50	398	38.8	147	45	203	3		
Cigarette smoking							32.912	1.09E-05
Not available	63	6.14						
No smoking (1)	719	70.08	18	372	278	51		
Former smoking (2)	47	4.58	0	12	26	9		
Smoking (3)	197	19.2	7	73	88	29		
Alcohol intake							31.42	2.51E-04
Not available	22	2.14						
No drinking (1)	858	83.63	21	436	332	69		
Once a week (2)	88	8.58	3	26	45	14		
Twice to three times a week (3)	33	3.22	1	7	17	8		
More than 3 times a week (4)	25	2.44	0	12	11	2		
Food shortage							3.3623	3.39E-01
Not available	58	5.65						
Yes (1)	73	7.12	1	29	34	9		
No (2)	895	87.23	24	439	351	81		
Meat Frequency							26.833	8.17E-03
Not available	1	0.10						
Everyday (1)	121	11.79	52	13	52	4		
Four or five times a week (2)	122	11.89	60	11	46	5		
Two or three times a week (3)	411	40.06	228	31	142	10		
No more than once a week (4)	317	30.90	122	38	150	7		
Never (5)	54	5.26	27	2	23	2		
Excretions Frequency							27.619	1.10E-03
Not available	3	0.29						
More than once a day (1)	96	9.36	33	17	44	2		
Once a day (2)	810	78.95	381	71	335	23		
Once every two to three days (3)	112	10.92	69	5	35	3		
Less than twice a week (4)	5	0.49	5	0	0	0		

Materials and Methods

Study Design and Ethics Statement

The majority of Chinese population is Han population. To identify potential molecular mechanisms associated with overweight and obesity, it is necessary to establish a China's Han population-based cohort to determine overweight and obesity, thus we established a large prospective population-based study in Xinmi, Zhengzhou in July 2017, and we randomly selected healthy subjects who have lived for a long time in Xinmi, Henan, China with at least three generations or more to participate in our study. This population represents a typical Han Chinese population which represents the standard sample of Chinese Han genes. Therefore, Xinmi is an ideal city, a district of Zhengzhou for our cohort research. Our study recruited 1026 Chinese

Han healthy people to investigate the miRNA expression levels in overweight and obesity in China. We collected blood samples from 1026 healthy Chinese adults in Xinmi and performed miRNA transcriptome and proteomics analysis of 1026 subjects to serve as a new framework for investigating new mechanisms of overweight- and obesity-related diseases. In this study, all the protocols, procedures and volunteer recruitment as described were approved by the Institutional Review Board (IRB)/Research Subject Review Board (RSRB) committee of the Zhengzhou University (approval number: ZZUIRB2021-144). All volunteers provided written informed consent. Our population-based study outline from the Han Cohort subjects briefly describes the study by investigating risk factors, biomarkers, and molecular mechanisms, and by developing intervention measures and novel therapies for chronic diseases in the future (Fig. 1).

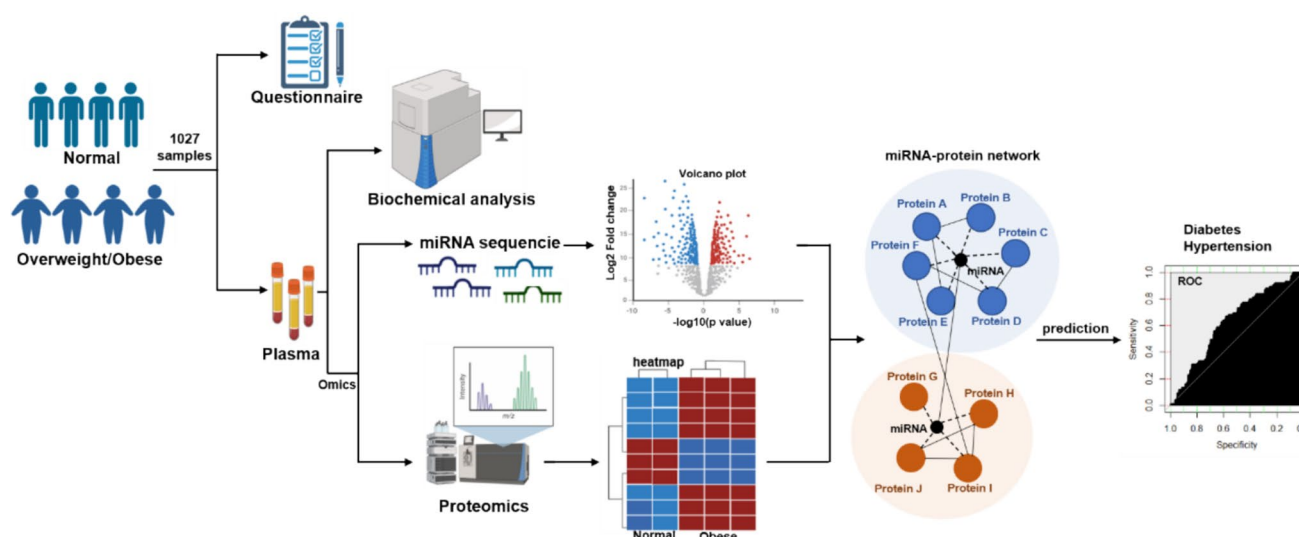


Fig. 1 Schematic representation of study design and concept of the obesity phenotype among Chinese Han adults. Blood samples of 1026 Chinese Han adults were collected in 2017 and plasma was

separated, then the plasma samples of 1026 subjects were used for microRNA sequencing and proteomics analysis. The miRNA-protein regulation networks about the obesity phenotype were indicated here

The Sample Collection of 1026 Chinese Han Adults

Blood samples (1026 samples) were randomly collected from Chinese Han adults. Plasma was then extracted from blood samples using a centrifuge and used for miRNA transcriptome sequencing and proteomics analysis. Data (1026 samples) on sex, age, cigarette smoking, alcohol intake, food scarcity, meat consumption, and excretion frequency were collected. Height and weight were measured, and cholesterol, high-density lipoprotein, low-density lipoprotein, triglycerides, and blood glucose were measured.

Phenotypic Acquisition and Measurement of Blood Pressure and Glucose

The participants sat and rested in a quiet environment for 15 to 20 min, were restricted from alcohol or substances that might alter blood pressure, and assumed a sitting position. The blood pressure of each subject was measured in their left and right arms, with the blood pressure cuff inserted about 10 cm above the antecubital fossa, at the same level as the heart. The average value of three measurements was taken. The venous blood test was performed in the hospital, and the subjects had fasted for more than 8 h before the blood samples were obtained to measure the fasting blood glucose. The median cubital vein or superficial blood vessels on the back of the hand were chosen. The blood glucose concentrations were measured by a fully automatic clinical biochemical analyzer.

Differentially Expressed miRNA Analysis

The plasma samples were sequenced using the Illumina NovaSeq 6000 System. Sequence data were subjected to alignments and annotations according to miRbase, a database for miRNA. The miRNA-seq count results were filtered out, and miRNAs with total counts in less than 10% of samples were removed. The differentially expressed miRNAs between the normal and the overweight/obese groups were examined using the linear contrast in the DESeq2 (Sundar et al. 2019). The threshold of differentially expressed miRNAs was $p < 0.05$ and an absolute value of $\log_2$ fold change (FC) > 0.3 . A volcano plot was generated using the *ggplot2* package in R 3.6.1 (Ito and Murphy 2013). A heatmap was plotted using the *pheatmap* package (<https://CRAN.R-project.org/package=pheatmap>) in R 3.6.1.

Identification of miRNA Signature for Predicting miRNAs Associated with Overweight/Obese

We analyzed the data from miRNA-seq results and identified the association of 21 differentially expressed miRNAs from these subjects with overweight/obese groups using miEAA (a miRNA enrichment analysis tool; <https://ccb-compute2.cs.uni-saarland.de/mieaa2/>) (Kern et al. 2020). The results from miEAA were further visualized using the *pheatmap* package in R 3.6.1.

Protein Mass Spectrometry, Quantification and Differentially Expressed Proteins Analysis

A threshold of 10% of the number of proteins detected in all samples was used to include samples for subsequent differential protein analysis, and 895 samples were ultimately retained. The 895 blood samples were pretreated to remove high-abundance proteins and interfering substances, and then the samples were subjected to trypsin enzymatic hydrolysis and peptide fractionation as described (Liu et al. 2017). The samples were further analyzed on a Thermo HFX MS (Thermo Fisher Scientific) interfaced with an EASY-nLC1200 LC system (Thermo Fisher Scientific). Raw files were searched against the human RefSeq protein database with Proteome Discoverer (Thermo Fisher Scientific, version 1.4) (Liu et al. 2017). The levels of proteins were estimated using the label-free, intensity-based absolute quantification (iBAQ) approach (Schwanhaussner et al. 2013). The iBAQ of each protein was normalized to the control by dividing the sum of all identified proteins. The matrix file integrated with all samples was filtered to remove proteins with summed counts in less than 10% of the samples, then was used to analyze differentially expressed proteins between the normal and overweight/obese groups using Wilcoxon rank sum test method. Different proteins were filtered by absolute values of Log_2 fold change (FC) > 0.5 and p -value < 0.05. Volcano and heatmap plots showed 230 differentially expressed proteins.

Pathway Enrichment Analysis to Obtain Protein Profiles Associated with Overweight and Obese Groups

The differentially expressed proteins were used to carry out pathway enrichment analysis using the *clusterProfiler* package in R 3.6.1 (Xu 2021). Default parameters (p -value cut-off, 0.05; q -value cut-off, 0.2; p -value adjustment method, Benjamini & Hochberg (BH)) were set for pathway enrichment analysis. Protein expression levels in the pathways were considered significantly different at p -value < 0.05 and q -value < 0.2.

Results

The Identification of Risk Factors Associated with Obesity Among Chinese Han Population

In this study, we classified 1026 samples into underweight (BMI < 18.5), normal ($18.5 \leq \text{BMI} < 25$), overweight ($25 \leq \text{BMI} < 30$) and obese groups (BMI ≥ 30) based on the BMI standard of the World Health Organization adult classification (Wildman et al. 2004). The analysis results showed

that among the Chinese Han adults in 2017, both males and females were mostly in the normal and overweight categories (Fig. 2a), and this overweight trend was similar in the follow-up survey results of 2020 (Fig. 2b). Importantly, we followed 596 volunteers and further compared the changes of BMI levels 3 years later (Fig. 2c). We also found that compared to female BMI levels, male BMI tended to increase more slowly after adulthood, and it continuously increased in women before the age of 56, especially between the ages of 40 to 50 (Fig. 2d). To explore the potential risk factors that affect BMI, we analyzed the correlation of sex, age, cigarette smoking, alcohol intake, experiences of food shortage, meat frequency and excretion frequency with BMI. The results showed that sex, age, cigarette smoking and alcohol intake were highly associated with BMI in Chinese Han population using the chi-square test (Supplement 1). Meat frequency and excretions frequency were also strongly correlated with BMI. These results displayed the status of obesity and living conditions associated with BMI in the Chinese population.

Identification of Differentially Expressed miRNAs that are Associated with Overweight and Obese Groups Among the Chinese Han Population

In this study, we analyzed whether specific miRNA expression levels were connected with overweight and obesity. For the follow-up analysis, 966 samples were divided into two groups based on BMI: normal and overweight/obese. We performed RNA sequencing of all 1026 plasma samples and analyzed the results to assess differential miRNA expression levels in the cohort. After quality control and standardization of raw count data, we ultimately obtained 639 miRNAs for differential expression analysis between the normal and overweight/obese groups. The volcano map showed that *hsa-miR-199b-5p* was the highest-expressed miRNA, while *hsa-miR-141-3p* was the lowest-expressed miRNA in overweight/obese group (Fig. 3a). The heatmap further showed that in the differentially expressed miRNAs, the overweight/obese samples had 11 miRNAs that were significantly lower than in the normal group, including *hsa-miR-26b-3p*, *hsa-miR-143-3p*, *hsa-miR-145-5p*, *hsa-miR-760*, *hsa-miR-205-5p*, *hsa-miR-654-5p*, *hsa-miR-1306-3p*, *hsa-miR-5010-5p*, *hsa-miR-655-3p*, *hsa-miR-196b-5p*, and *hsa-miR-141-3p*. In contrast, *hsa-miR-199b-5p*, *hsa-miR-548a-5p*, *hsa-miR-548b-5p*, *hsa-miR-362-5p*, *hsa-miR-181c-3p*, *hsa-miR-664b-5p*, *hsa-miR-148a-5p*, *hsa-miR-885-5p*, *hsa-miR-18a-3p*, *hsa-miR-590-3p*, and *hsa-miR-450b-5p* were expressed significantly higher in the overweight/obese group than in the normal group (Fig. 3b).

To investigate whether these differentially expressed miRNAs are involved in the obesity-related metabolic pathway, we used miEAA, a web-based miRNA

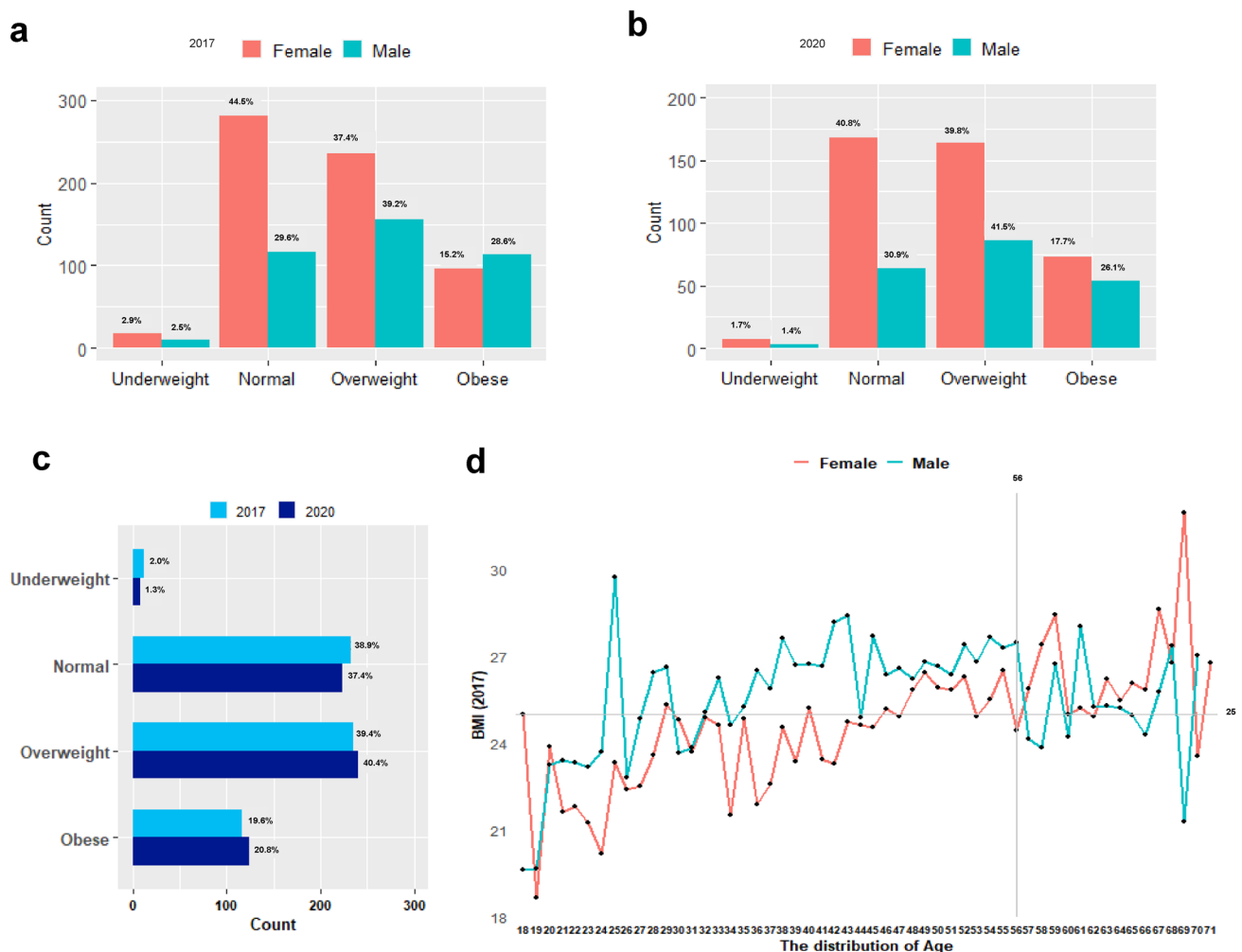


Fig. 2 The risk factors associated with obesity among Chinese Han adults. **a, b** The distribution of obesity levels (underweight, normal, overweight, and obesity) among Chinese Han adults in 2017 (female=631, male=395) and 2020 (female=412,

male=207). **c** The obesity distribution comparison of 596 Chinese Han adults in 2017 and 2020. **d** The BMI levels of men and women at different ages in 2017 ($n=1026$)

analysis application, to perform gene set enrichment analysis (GSEA) of miRNAs. We found that these differentially expressed miRNAs were associated with the TGF-beta signaling pathway, lipid metabolism, and insulin-related pathways (Fig. 3c). We further analyzed the expression levels of *miR-143-3p* and *miR-885-5p* in plasma samples of 10 normal and 10 obese Chinese Han adults using a different set of samples, the results confirmed that the expression levels of *miR-143-3p* were significantly lower, and the levels of *miR-885-5p* were slightly higher in the obese samples than in the normal group, which is consistent with the results of differential miRNA analysis (Fig. 3d). In summary, we found that the levels of a few circulating miRNAs were dysregulated in overweight and obese Chinese population, and these miRNAs are associated with the TGF-beta signaling and insulin pathways.

Relative Scores of Different miRNAs used to Identify miRNA Signature for Obesity to Measure the Degree of Obesity

We found 21 miRNAs that were significantly differentially expressed between the normal and the overweight/obese groups and further classified them using $\log_2$ fold change. To investigate the relationship between 21 differentially expressed miRNAs and degree of obesity, we first established a scoring procedure using the 21 differentially expressed miRNAs to assess the degree of obesity of all 996 samples. The specific grading scales are listed below: (1) The specific miRNAs in the sample are among the most highly expressed in the overweight/obese group. If the expression levels in a sample were lower than the median value of all samples, the score of this sample would be -1,

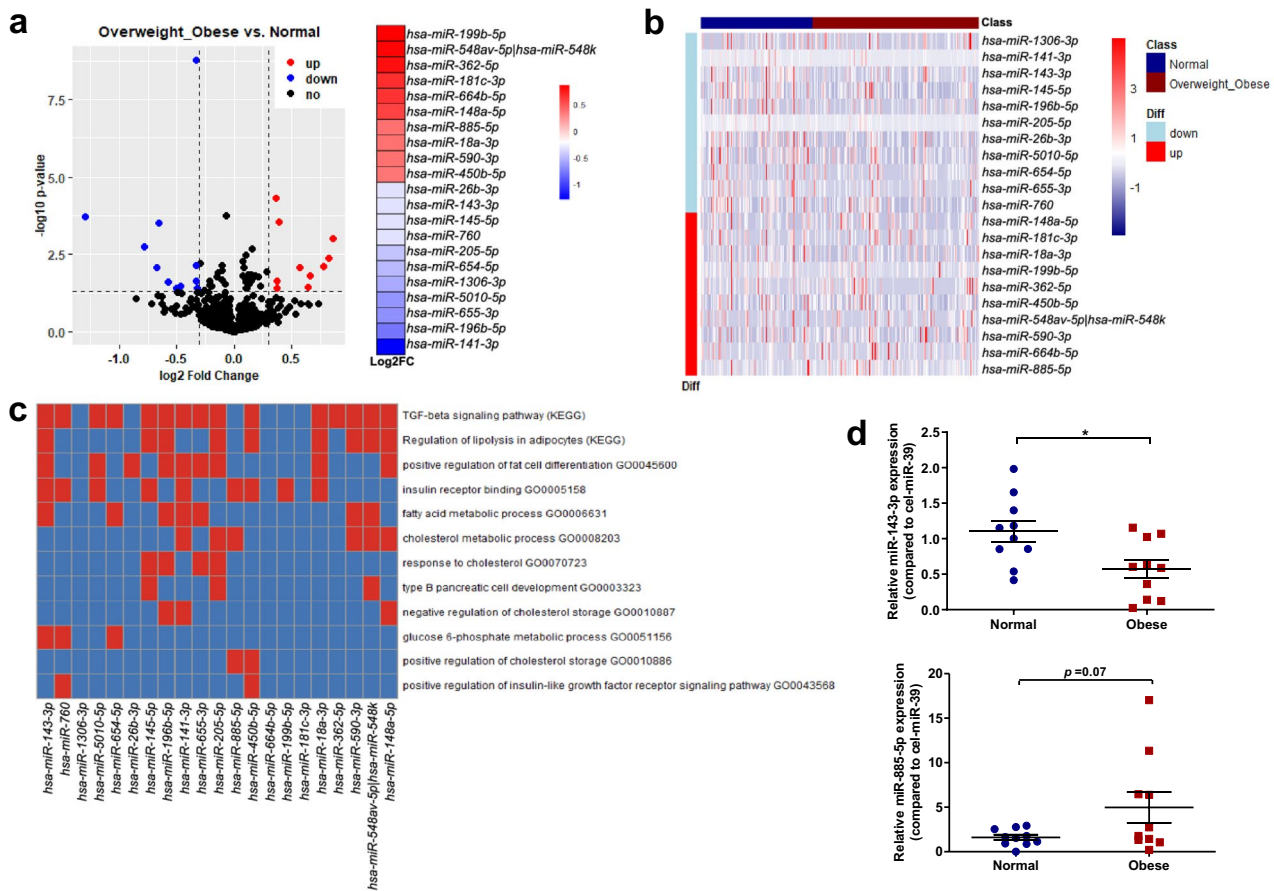


Fig. 3 Identification of differentially expressed miRNAs in normal and overweight/obese Chinese Han population. **a** Differentially expressed miRNAs between normal and overweight/obese groups (left: red indicates up-regulated miRNAs, blue indicates down-regulated miRNAs, and black indicates no differential expression; right: 21 differentially expressed miRNAs sorted by log₂ fold change). **b** Heatmap of 21 differentially expressed miRNAs among

normal and overweight/obese groups. **c** The most related 12 signaling pathways of differentially expressed miRNAs (red indicates miRNAs are highly involved in the corresponding pathway). **d** Expression levels of *hsa-miR-143-3p* in 10 normal and 10 obese samples of Chinese Han Adults; *cel-miR-39* was used as a control miRNA. *indicates significant difference at p -value < 0.05

whereas it would be enhanced by 1 if the levels were higher. (2) The miRNA in the sample is among the low-expressed miRNAs in the obese group. If its expression levels in a sample were lower than the median value of all samples, the score of that sample would be increased by 1, and vice versa. (3) We summed the scores of 21 miRNAs in all samples to determine the final scores of each sample (Fig. 4a). According to the relative scores and results, we found that as the scores were increased, the levels of BMI gradually increased, especially when the scores were greater than 1, the BMIs were higher (Fig. 4b). We next defined the samples with scores less than 0 as the low score group, and those with a score greater than 0 as the high score group, then we further compared the levels of BMI and obesity-associated clinical indicators between the two groups. The results of the statistical test showed that the higher score group had significantly higher BMI and levels of cholesterol,

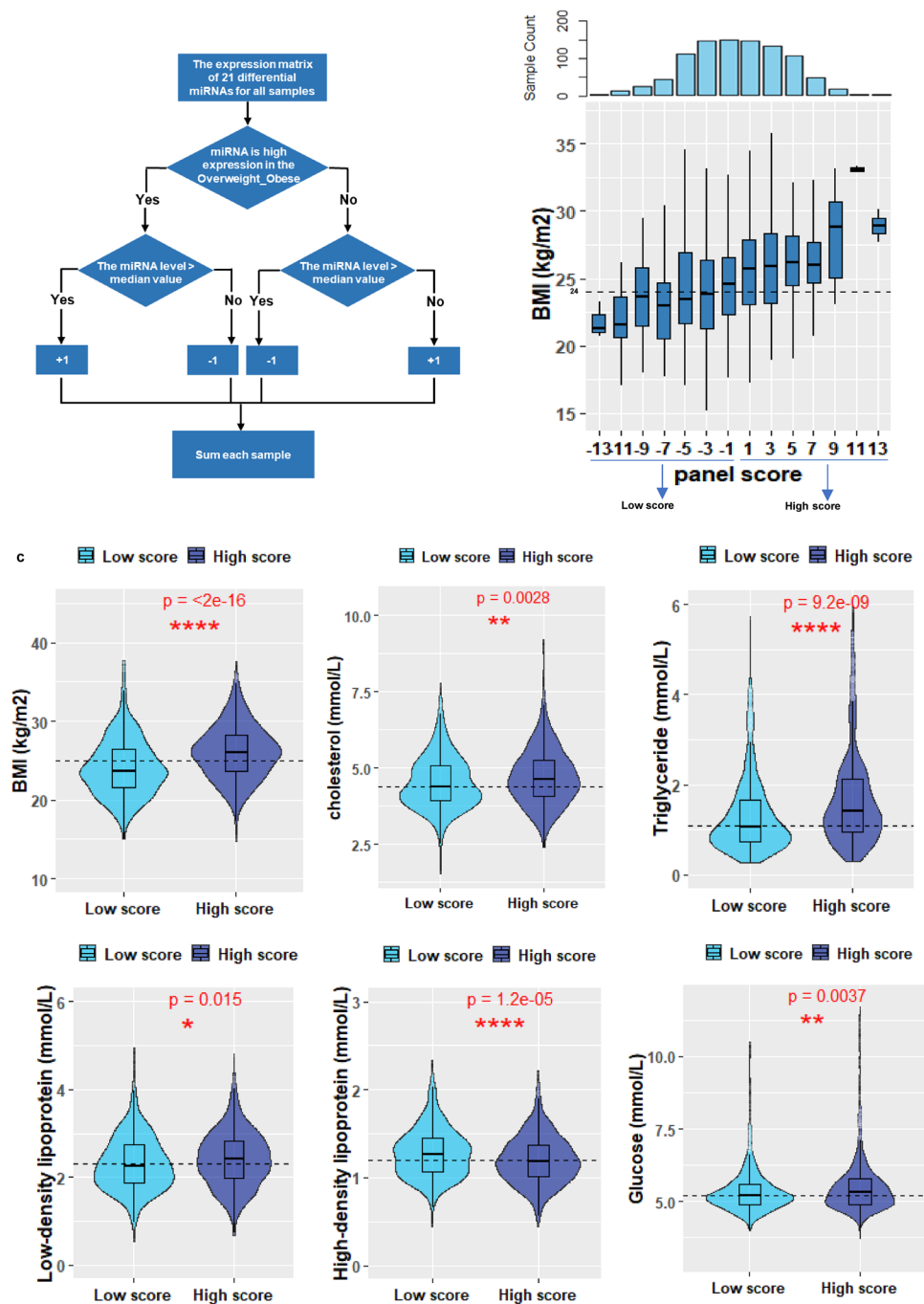
triglycerides, low-density lipoprotein, and glucose, whereas high-density lipoprotein levels were significantly lower in the high score group (Fig. 4c). These results indicated that there was a strong correlation between the 21 differentially expressed miRNAs and obesity as well as obesity-related high blood lipids and blood glucose.

Differentially Expressed Protein Panels and miRNA Direct Targets Associated with Overweight and Obese Groups

To further explore the mechanisms of miRNA anomalies that contribute to metabolic disorders in the Chinese Han population, we performed proteomic analysis on 996 samples and collected data from 895 subjects. First, we compared protein expression levels in normal and overweight/obese groups. The volcano map showed that 230 proteins

Fig. 4 Relative miRNAs scores for obesity difference were a reasonable method to measure the different degrees of obesity.

a The flow chart of scoring of each sample according to 21 differentially expressed miRNAs. **b** The distribution of sample scores (up) and BMI levels of sample groups with different scores (down). **c** The levels of BMI, cholesterol, triglyceride, low-density lipoprotein, high-density lipoprotein and blood glucose among low score and high score groups. The low score group includes samples whose scores were lower than 0, and the high score group includes samples whose scores were higher than 0. * indicates significant difference at p -value < 0.05 , ** indicates significant difference at p -value < 0.01 , and **** indicates significant difference at p -value < 0.0001



were significantly differentially expressed between the two groups (Fig. 5a). The heatmap revealed that 128 proteins were significantly overexpressed and 102 proteins were significantly underexpressed in the overweight/obese group (Fig. 5b). After sorting 230 differentially expressed proteins by Log fold change ($\log_2$ FC), we found that LLGL Scribble Cell Polarity Complex Component 1 (LLGL1) was the highest-expressed protein. Besides, Dehydrogenase/Reductase 1 (DHRS1) was the lowest-expressed protein in the overweight/obese group (Fig. 5c). Furthermore, these

differentially expressed proteins were subjected to pathway enrichment analysis, and the results showed that these proteins were mainly enriched in the RNA catabolic process, which implied that these proteins may cause dysregulation of intracellular signaling pathways by affecting the RNA process (Fig. 5d). In order to find the cardinal proteins that affect RNA catabolic process, we further analyzed the results of pathway enrichment and visualized them using a Chord diagram, and the results showed that SNW1, EXOSC3, SNRPA, CIRBP, PSMO4, HNRNPU, GEMIN5, PNN,

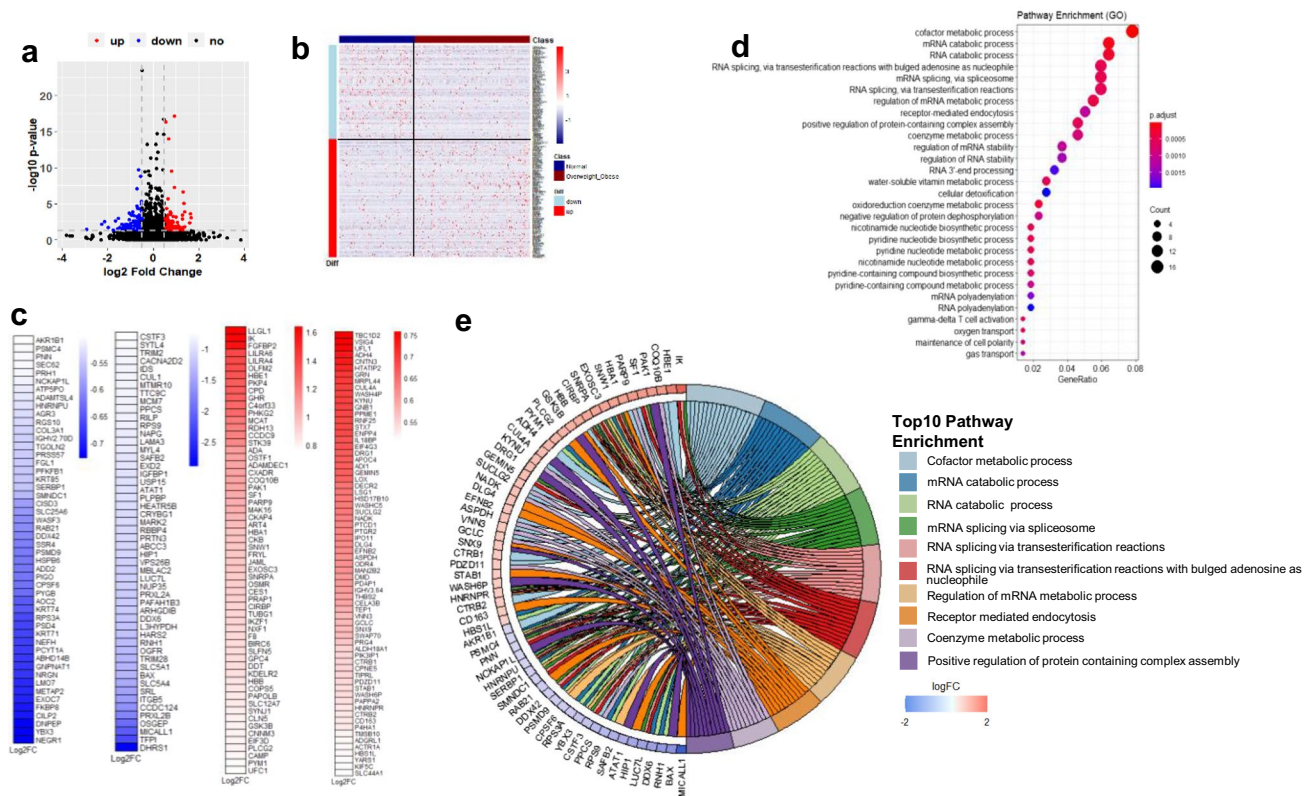


Fig. 5 Differentially expressed proteins between normal and overweight/obese Chinese Han population. **a** The differentially expressed proteins between the normal and overweight/obese group (red indicates up-regulated proteins, blue indicates down-regulated proteins, and black indicates no differential expression). **b** The heatmap of 230 differentially expressed proteins among the normal and overweight/obese group. **c** The differentially expressed proteins were sorted

by log₂ fold change (log₂ FC). Red represents up-regulated proteins, and blue represents down-regulated proteins. **d** GO pathway enrichment analysis for differentially expressed proteins. The sizes of the dots represent the number of enriched genes, and color indicates the adjusted *p*-values. **e** Chord diagram showed 10 biological processes and key proteins from GO analyses results

SMNDC1, and SERBP1 proteins were involved in multiple pathways (Fig. 5e). In these analyze, we found a series of new proteins that were highly correlated with obesity.

Identification of miRNA Signature Associated with the Obesity Phenotype of the Chinese Population

To further investigate the miRNAs-protein regulation network, three databases (TargetScan, miRWalk 2.0, and miRDB) were used to identify regulating targets of the 21 differentially expressed miRNAs. We then identified the aggregated 5400 miRNA targets and 230 differentially expressed proteins and found that 54 proteins overlapped (Fig. 6a). These overlapping proteins were enriched using Metascape, a web-based program, and we found that the top two significant enrichment pathways were mRNA surveillance and regulation of receptor-mediated endocytosis (Fig. 6b). In order to clarify the regulatory relationship between miRNAs and proteins, we developed a miRNAs-Proteins-Biological functions regulatory network

using Cytoscape, a network visualization tool, and uncovered several key regulation pathways, such as *hsa-miR-548av-5p* and *miR-548k* that regulated PNN to affect the mRNA surveillance pathway, which regulated PLCG2 to affect the positive regulation of receptor-mediated endocytosis and the AGE-RAGE signaling pathway in diabetic complications, and regulated STK39 to affect the negative regulation of glycogen biosynthesis. *Hsa-miR-199b-5p* regulated HIP1 to affect the process of positive regulation of receptor-mediated endocytosis, and EXOSC3 to affect the process of RNA phosphodiester bond hydrolysis (Fig. 6c). To summarize, we discovered new control networks of miRNA-protein changes in Chinese Han overweight/obese individuals.

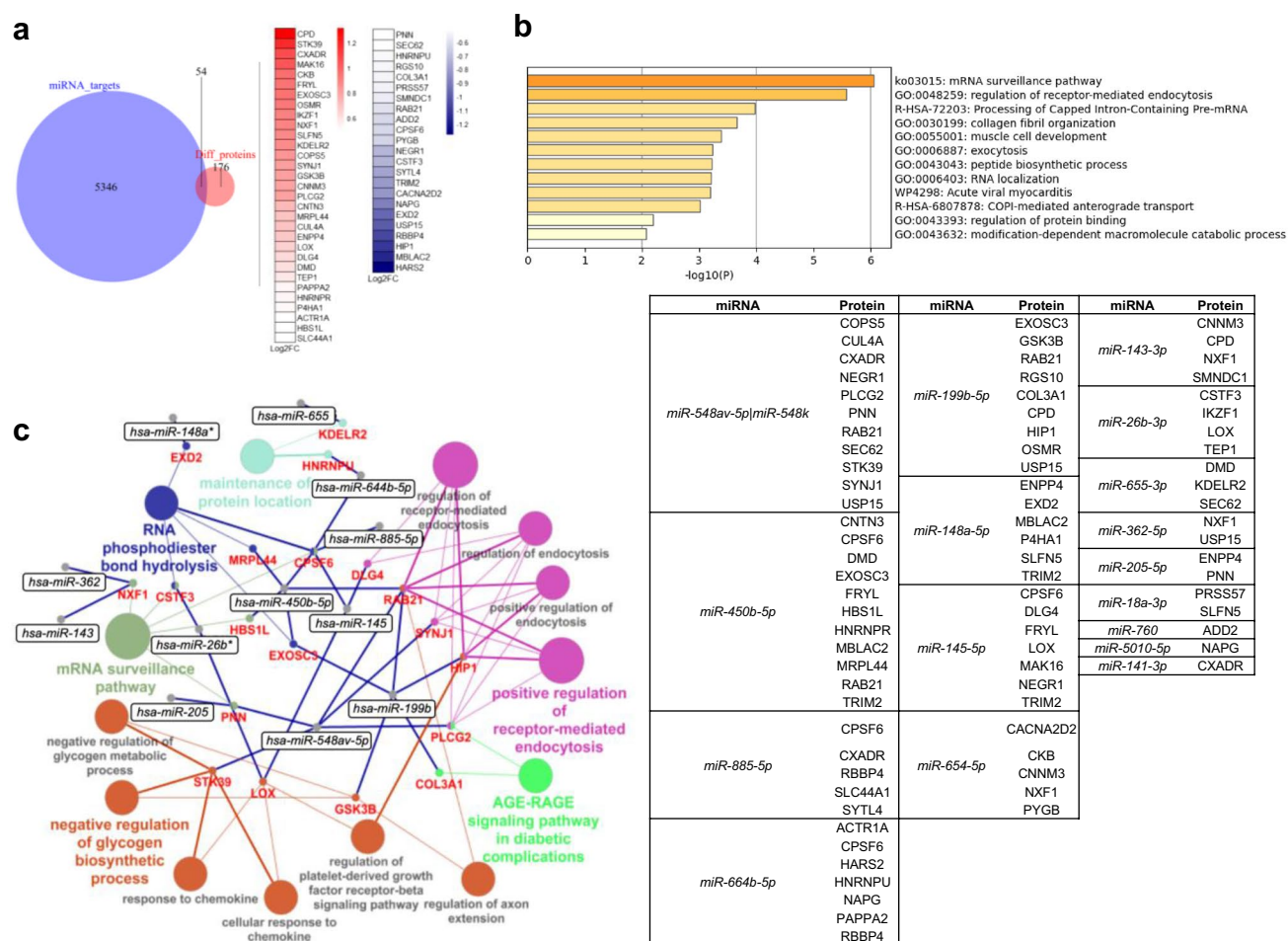


Fig. 6 The miRNAs-Proteins regulatory network was reshaped in the obese phenotype. **a** The Venn diagram showed the overlap between the targets of 21 differentially expressed miRNAs and differentially expressed proteins. The 54 overlapping proteins were sorted

by $\log_2$ FC. **b** The pathway enrichment analysis of 54 overlap proteins. **c** The regulation network diagram of the relationship between miRNAs, proteins, and biological functions. The same color represents similar functional modules

Combination of miRNA Signatures and Protein Panels Associated with Overweight/Obese in the Chinese Han Population which are Correlated with Blood Glucose and Blood Pressure

Obesity is the cause of numerous diseases, including high blood pressure and diabetes. We then determined whether there is a link between obesity-related miRNAs and proteins, as well as blood pressure and blood glucose levels, respectively, in Chinese Han population. Therefore, we first performed a correlation analysis of 21 obesity-related miRNAs with the mean values of three systolic and diastolic blood pressures, respectively, and found that *hsa-miR-143-3p*, *hsa-miR-205-5p*, *hsa-miR-145-5p*, *hsa-miR-141-3p*, *hsa-miR-760* and *hsa-miR-885-5p* levels were associated with both systolic blood pressure and diastolic blood pressure (Fig. 7a). The levels of obesity-related proteins: NEGR1, CACNA2D2, PYGB, CUL4A and CNTN3 were associated

with both systolic and diastolic blood pressure (Fig. 7b), and CACNA2D2 levels may be used to independently predict the risk of progression of home hypertension based on longitudinal study of 403 Japanese subjects (Watanabe et al. 2010). Furthermore, we assessed the efficacy of blood pressure-related miRNAs and proteins in identifying hypertensive subjects from healthy participants. We characterized the hypertensive group as having a systolic blood pressure of at least 140 mmHg and/or diastolic blood pressure of at least 90 mmHg two times. In addition, we included 295 patients with hypertension. Receiver operating characteristic (ROC) curve analysis revealed that *hsa-miR-143-3p*, *hsa-miR-205-5p*, *hsa-miR-145-5p*, *hsa-miR-141-3p*, *hsa-miR-760* and *hsa-miR-885-5p* levels had ROC values (0.597) for discriminating hypertension from healthy controls (Fig. 7c). The combination of miRNAs and hypertension-associated proteins achieved an ROC values of 0.63 (Fig. 7d). The results indicated that combining miRNAs and

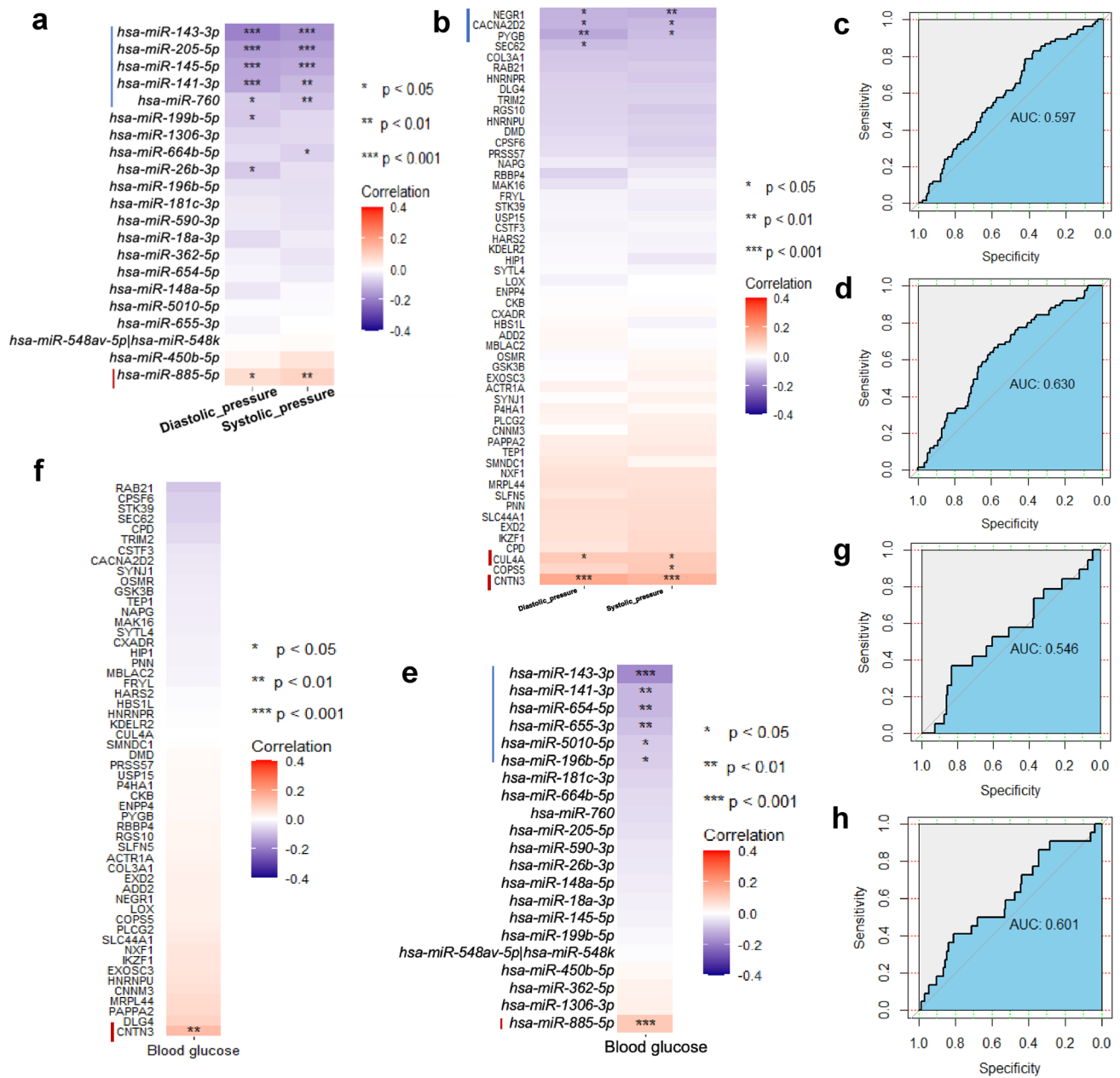


Fig. 7 Combination of several miRNAs and proteins have better prediction values for overweight/obese Chinese Han population associated with blood glucose, blood pressure. **a** Pearson's correlation analysis between 21 obesity-related miRNAs and diastolic, and systolic blood pressure. **b** Pearson's correlation analysis between obesity-related proteins and diastolic and systolic blood pressure. **c** The ROC curve of blood pressure-related miRNAs in distinguishing hypertension from healthy controls. **d** The ROC curve of blood

pressure-related miRNAs and proteins in distinguishing hypertension from healthy controls. **e** Pearson's correlation analysis between 21 obesity-related miRNAs and blood glucose. **f** Pearson's correlation analysis between obesity-related proteins and blood glucose. **g** The ROC curve of blood glucose-related miRNAs in distinguishing hyperglycemia from healthy controls. **h** The ROC curve of blood glucose-related miRNAs and proteins in distinguishing hyperglycemia from healthy controls

proteins improved the accuracy of hypertension prediction. Similarly, the correlation between obesity-related miRNAs and fasting blood glucose was investigated. The levels of *hsa-miR-143-3p*, *hsa-miR-141-3p*, *hsa-miR-654-5p*, *hsa-miR-655-3p*, *hsa-miR-5010-5p*, *hsa-miR-196-5p* and *hsa-miR-885-5p* were significantly different (Fig. 7e). In the

analysis of obesity-related proteins, only CNTN3 levels were associated with blood glucose (Fig. 7f). We investigated the effectiveness of blood glucose-related miRNAs and proteins in differentiating hyperglycemic individuals from healthy individuals. We identified 62 hyperglycemic subjects with fasting blood glucose levels more than 7.0 mmol/L. ROC

curve analysis revealed that blood glucose-related miRNAs produced ROC values of 0.546 for distinguishing hyperglycemia from healthy controls (Fig. 7g). The combination of miRNAs and hyperglycemia-related protein CNTN3 levels yielded an ROC values of 0.601 (Fig. 7h). The findings also suggested that the combination of miRNAs and protein may be effective in hyperglycemia prediction. Together, these finding suggested that a combination of obesity-related miRNAs and proteins could be used as biomarkers associated with disease status (hypertension and hyperglycemia) in the Chinese Han population.

Discussion

Overweight and obesity are described as abnormal or excessive fat buildup that poses a risk to human health, and BMI is the most direct way to evaluate overweight and obesity. Weight and obesity rates continue to rise among adults and children around the world. In 1985, the National Institutes of Health (NIH) stated that BMI higher than 27.8 for men and 27.3 for women were strongly associated with high health risk. In recent years, the standard has been further categorized, with a BMI equal to or higher than 25 in both men and women being considered overweight, and a BMI equal or higher than 30 was considered obese, the BMI values for standard body weight are $18.5 \leq \text{BMI} < 25$. In 1026 samples of China Han population, we found that BMI values were strongly correlated with sex, age, cigarette smoking, alcohol intake, meat frequency, and excretions frequency. Based on China's adult BMI classification, we divided 1026 samples into four groups: underweight, normal, overweight, and obese. We discovered that the overweight group and obese group had higher levels of cholesterol, triglycerides, low-density lipoprotein, glucose, and lower levels of high-density lipoprotein.

Recently, RNA sequence analysis from blood, leukocytes, plasma, saliva, and serum revealed that specific expression levels of miRNAs, tRNAs and piRNAs may be employed as biomarkers of overweight or obesity (El-Mogy et al. 2018). To date, no studies have been performed to determine whether miRNA expression levels in blood are associated with overweight or obesity in the Chinese population. We predicted that the expression levels of several miRNAs and their downstream target genes can predict overweight and obesity in Chinese population. To verify this theory, we first performed a comparative analysis of miRNAs and their target genes based on RNA sequencing and proteomics analysis of peripheral blood from Chinese Han adults that were differentially expressed in overweight and obese people versus normal weight individuals.

There had been reported that *miR-155* was a known pro-inflammatory miRNA and was significantly upregulated

in microvesicles that were secreted by adipocytes derived from obese mice as compared with lean mice (Kurowska-Stolarska et al. 2011; Zhang et al. 2016). The expression levels of *miR-802* were increased in the pancreatic islets of obese mouse models, and induced the impaired insulin transcription and secretion via Foxo1 (Zhang et al. 2020). *MiRNA-10a-5p* modulated the diurnal rhythm of *Lachnospiraceae* and its metabolite butyrate, then further caused HFD-induced glucose intolerance and insulin resistance in High-Fat Diet-Fed Mice (Guo et al. 2020). In addition, *miR-let-7* targeted multiple genes containing SNPs associated with type 2 diabetes and control of fasting glucose, and the *miR-let-7* pathway was a central regulator of mammalian glucose metabolism (Zhu et al. 2011).

We found that levels of 21 miRNAs were significantly differentially expressed between normal and overweight/obese groups. The overweight/obese people had significantly lower expression levels of *hsa-miR-26b-3p*, *hsa-miR-143-3p*, *hsa-miR-145-5p*, *hsa-miR-760*, *hsa-miR-205-5p*, *hsa-miR-654-5p*, *hsa-miR-1306-3p*, *hsa-miR-5010-5p*, *hsa-miR-655-3p*, *hsa-miR-196b-5p* and *hsa-miR-141-3p*. In contrast, levels of *hsa-miR-199b-5p*, *hsa-miR-548av-5p*, *hsa-miR-548 k*, *hsa-miR-362-5p*, *hsa-miR-181c-3p*, *hsa-miR-664b-5p*, *hsa-miR-148a-5p*, *hsa-miR-885-5p*, *hsa-miR-18a-3p*, *hsa-miR-590-3p*, and *hsa-miR-450b-5p* had higher expression levels in the overweight/obese samples. These differentially expressed miRNAs are mainly involved in TGF-beta signaling pathway, lipid metabolism and insulin related pathways. Finally, we demonstrated that the panels of differently expressed miRNAs were related to obesity indicators and these circulating miRNAs were involved in the body's metabolic network, which may lead to obesity-related diseases. MiRNAs are known to play an essential role in regulating gene expression in several different physiological and pathophysiological conditions (Nana-Sinkam et al. 2017; Yanez-Mo et al. 2015). Additionally, it is reported that dysregulations of various miRNAs in the adipogenic, adipocyte differentiation, glucose and lipid metabolism in overweight and obesity (Castano et al. 2018; Hernandez-Diazcorder 2019; Shi et al. 2016a, 2016b; Zhu et al. 2011). In this study, we found that miRNAs were among the major biotypes distributed from peripheral blood that were differentially expressed in overweight people compared to normal weight people.

The regulatory functions of miRNAs were usually through their downstream target genes (Kabekkodu et al. 2018; Liu et al. 2014). Overweight and obesity are main risk factors for diabetes, and insulin resistance is one of the important causes of type 2 diabetes (Kao and Sabin 2016). To further investigate the mechanisms of the panel of differently expressed miRNAs in regulating the overweight and obesity, we performed proteomic analysis to analyze qualified samples of 895 subjects among 996 normal and

overweight samples, all the target genes of above differentially expressed miRNAs were analyzed, and we found that the target proteins were mainly enriched in RNA catabolic process. These results suggest that these proteins may cause dysregulation of intracellular signaling pathways by affecting RNA processing. The panels of differently expressed miRNAs and their target proteins formed a complex regulatory network to regulate the developing process of overweight and obesity in Chinese population, which warrants further investigation. Additionally, we further verified the expression levels of *miR-143-3p* and *miR-885-5p* in plasma samples of 10 normal and 10 obese people, the results showed that the expression levels of *miR-143-3p* were significantly lower and *miR-885-5p* levels were higher in the obese samples, which is consistent with our analysis results of differential miRNA analysis. It was also reported that the *miR-143-3p*-IGF2R signaling pathway may be a potential target for the treatment of obesity-associated insulin resistance (Xihua et al. 2019). *MIR-143-3p* regulated the preadipocyte proliferation and differentiation by targeting MAPK7 (Zhang et al. 2018a, 2018b). In our study, we found that *miR-143-3p* had high correlation with the obese associated diseases such as high blood glucose, blood pressure and blood lipids.

This study shows association of miRNA signature and protein targets with overweight/obese in the Chinese population. The levels of these miRNAs and target protein panels may be potential biomarkers and prevention targets of overweight and obesity in the Chinese population. Future studies are needed to identify more biomarkers and therapeutic targets in the follow-up investigation.

In summary, we have taken efforts to isolate and characterize plasma from Chinese population. We employed RNA sequencing analysis to identify the specific miRNAs, and targets of miRNAs that are significantly different between the overweight/obese subjects and normal weight people. Proteomic analysis also identified a number of targets that may play crucial roles in overweight, obesity and several overweight associated diseases. We found 21-miRNAs signature (10 upregulated and 11 downregulated miRNA clusters) that are associated with obesity and obesity-related diseases; and identify their downstream targets and protein panel that were highly associated with the overweight and obesity in the Chinese population.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s43657-025-00259-0>.

Authors' Contributions BHJ, LJ, and SJW designed the project and provided resources for the study. YXX and JGQ performed the experiments, conducted data analysis, and wrote the manuscript. For the replication and meta-analysis, CD, LMS, and ZHY performed some experiments. All authors read and approved the final version of the manuscript.

Funding The project was supported in part by the Shanghai Municipal Science and Technology Major Project (Grant No. 2017SHZDZX01) and by National Natural Science Foundation of China (Grant No. 82073393).

Data Availability The data analyzed during the current study are available from the corresponding authors on reasonable request.

Declarations

Conflicts of Interests We declare no conflicts of interests.

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